A Mini Project report on

Audio-Visual Action Recognition using Cricket Dataset

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Submitted By

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CERTIFICATE

This is to certify that the project entitled **"Audio-Visual Action Recognition using Cricket Dataset"** is a bonafide work carried out by the student team Sudeep Sanshi (01FE23BCI101), Guru Bandathar (01FE23BCI120), Chetan Honnalli (01FE23BCI108), and Nandeesh I B (01FE23BCI092) in partial fulfillment of the requirements for the completion of the 5^{th} semester Mini Project during the academic year 2025–2026. The project report has been approved as it satisfies the academic requirements with respect to the project work prescribed for the above said course.

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ABSTRACT

Action recognition in sport videos is challenging owing to the interaction of audio and video characteristics resulting from motion. The multimodal action recognition system developed in this project uses ResNet18 for video feature extraction and YAMNet for audio feature extraction as the task is related to action recognition in cricket videos. The dataset was developed in house using videos obtained from YouTube, and each action was marked within its precise time boundary and then split into labeled pieces based on Four runs, Six runs, Wicket, and Others categories as they all require equal samples and are divided into train, validation, and testing in the ratio of 80:10:10 for proper learning and development. For the audio and visual data to be combined efficiently, an "attention-based fusion mechanism using the Deep Neural Network (DNN)" technique can be employed. Using the attention mechanism, the audio and visual prediction outputs can be given appropriate weights based on the relevance of audio and visual in the context of the respective actions. The fused representation is then used for final classification. Extensive experiments were conducted using early, late, gated, and attention fusion strategies, with attention fusion achieving the highest and most stable accuracy of 86%. Performance evaluation using confusion matrices and classification reports demonstrates balanced class-wise results and reduced misclassification compared to traditional fusion methods. The results confirm that attention-based multimodal fusion with pretrained semantic audio embeddings significantly improves action recognition performance in sports video analysis.

Keywords : *Multimodal Action Recognition, Audio-Visual Fusion, Attention-Based Fusion, Deep Neural Network (DNN), ResNet18, YAMNet, Sports Video Analysis, Cricket Event Classification, Custom Dataset, Confusion Matrix Analysis.*

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INTRODUCTION

The recognition of sports actions in video streams is a challenging and important problem in the field of computer vision and multimedia analysis. With the rapid growth of sports broadcasting and digital media platforms, there is an increasing demand for automated systems that can accurately interpret and classify key events from sports videos. For instance, event recognition has a vital application area in match analysis, highlight reels, performance analytics, and decision support systems [1][2][3]. Action classification systems form the backbone of such technologies by enabling machines to understand and label meaningful game events from video data.

In the past, action recognition in the game of cricket had traditionally been conducted mostly using visual information. There had been the analysis of a sequence of frames that aided in the recognition of the patterns of action that took place using the services of convolutional networks. It is actually true that the purely visual action recognition approach had faced immense difficulties in actual broadcasts of the game that contained fast camera movements, occlusions, viewpoints, as well as a high amount of visually similar action. An action like Four and Six had a nearly similar visual representation.[3][4][5]

To address these issues, current approaches have started to move towards multimodal learning methods that leverage other modalities along with visual inputs. Audio is a great additional source for cricket video analysis as the crowd reaction while a six is hit or a sudden audio dynamic change when a wicket falls gives some excellent semantic clues about what is happening.The explosive crowd reaction associated with a six or a dramatic audio dynamics change associated with a wicket can assist with ambiguity between different actions that have a high visual similarity. A system that observes both audio and visual features can more effectively classify cricket activities than a system that observes only visual features by emphasizing both visual and audio cues at critical stages of cricket activities that result in a six or a wicket.

In the project work, a multimodal audio-visual action classification approach is designed for identifying the key events in the cricket video clips. For extracting the high-level visual features, deep learning techniques like the usage of ResNet18, R3D-18, and C3D models are performed that focus on the spatial as well as spatiotemporal patterns of the player actions.

Simultaneously, various audio feature extraction methods like YAMNet, VGGish, CLAP, and MFCC are applied that emphasize the audio patterns like the reaction of the audience, the commentary level, as well as the collision sound of the bat & ball.[6][7][8]

To achieve the fusion of information from both modes, different fusion modes are assessed for this project, including early fusion, late fusion, gated fusion, and attention-based fusion. Early fusion involves the fusion of audio and images at the feature representation stage, while late fusion is the combination of the predictions from independent unimodal classifiers. Gated fusion is dynamic, controlling the weightage for each mode, while for attention-based fusion, the model gets trained for focusing on the most representative audio-visual features for each clip. The output from this fusion is then classified using a customized neural network model, with the goal of categorizing the output into one of the four classes representing different actions performed during a cricket match: Four Runs, Six Runs, Wicket, and Others. The project is evaluated using its performance metrics, including accuracy, precision, recall, F1-score, and confusion matrix analysis on a customized dataset for actions from videos used within the project.[9][10][11]

1.1 Preamble

Action recognition in sport-related video content can be described as the process of extracting significant events based on complex motion patterns in the video content. For instance, in cricket-related video content, action recognition can be considered even more complex as there exist fast camera movements, many changes in the video content, similar motion in various actions, and various coverage conditions. It can easily result in ambiguity based on vision alone for actions like four and six runs that could have almost similar vision characteristics.

Audio information acts as additional context that proves useful in supplementing the action recognition in cricket video analyses. The audience reaction, sound of bat–ball collision, and change in commentary loudness can prove helpful in identifying events despite confusing visual information. By combining both the visual and audio aspects of the given information, it is possible to distinguish between similar events in the cricket video analysis.[12][13][14]

These difficulties can be addressed with the adoption of a multimodal action recognition approach that includes the analysis of audio and video simultaneously. Features can be extracted from the video using deep learning architectures such as ResNet18, R3D-18, and C3D, which tend to capture the spatial and spatiotemporal details of cricket actions effectively. Meanwhile, audio features can also be extracted using methods such as YAMNet, VGGish, CLAP, and MFCC, which allow the process to capture semantic as well as acoustic aspects of cricketing actions.[15][16][17]

The extracted audio and vision features are combined using various fusion techniques such as early fusion, late fusion, gated fusion, and attention-based fusion to assess their efficacy systematically. The various fusion techniques allow flexible weighing and accenting of modality-related information, thereby generating more informative features. The fused features are then passed through a custom classification network to predict cricket-related activities like Four Runs, Six Runs, Wicket, and Others.

The proposed approach will be trained on a specially curated dataset of a cricket action that has been extracted using real recorded videos on TV. Through the synergy created between the usage of various transfer learning models and the backdrop of sophisticated multimodal learning models, the aim of the current project is to make robust models that can recognize a cricket action.

1.2 Motivation

Audio-Visual Action Recognition (AVAR) is a recent branch of multimodal learning that tries to identify actions by analyzing the visual and auditory features of the video together. In the context of analyzing cricket videos, action recognition is a challenging job because of the fast motion of the camera, the editing patterns of different channels, similar events in the video, and varying environments in which the video is recorded. The conventional action recognition systems that work solely with visual information fail to identify four runs and six runs in a cricket match because they have almost similar visual patterns.

Audio signals contain vital contextual information, which can prove quite helpful in comprehending cricket events. Crowd response, bat and ball collision sounds, and sudden variations in commentary loudness can turn out to be good indicators for cricket events in cases where the visual information is not clear or partially obscured. Synergistic analysis of audio and visual information can prove quite helpful in distinguishing similar events in action recognition systems.

Despite impressive advances in deep learning for action recognition, existing works primarily focus on generic benchmark datasets and unimodal or limited multimodal settings. There is a dire shortage of sport-specific datasets (e.g., cricket), and a broadcast cricket video presents some challenges: noisy audio, temporal misalignment of modalities, and domain-specific action patterns that standard benchmarks fail to provide ample representation for. This motivates the collection of a custom cricket action dataset and the exploration of tailored multimodal fusion strategies.

The main aim of this venture is to comprehensively assess several audiovisual fusion strategies, such as early fusion, late fusion, gated fusion, and attention-based fusion, using various pretrained audio and visual models. In this way, the objective of the contribution is to cast light on the importance of modalities in classification tasks and, in the process, establish an effective multi-modal solution for accurate action recognition in the context of cricket.

1.3 Literature Study

Sports action recognition using videos is a popular research endeavor because of its applications in sports analytics analysis, automatically generating sport highlights, summarizing sport broadcasts, among other decision-support services. The earlier works focused on the visual aspect of action recognition in sport videos.

But with advancements in deep learning, convolutional and spatiotemporal neural networks made great progress in sport actions recognition. Methods like C3D, I3D, and then 3D CNN became successful for their ability to learn spatial and temporal features jointly from imagery data. For cricket and other grounds-based games, there has been considerable work in applying vision-based deep learning for tasks such as shot identification, bowling action identification, and segmenting events. Unfortunately, pure vision-based methods cannot differentiate between two similar events, including four and six in cricket, which share common motion characteristics and lack context.

Recent approaches for the multimodal action recognition problem concentrated on deep fusion approaches for the audio-visual modality. The early fusion approaches attempt to combine the audio and video features at the representation level by simply stacking them, whereas the late fusion approaches attempt to combine the predictions for each modality independently at the decision level. The latest approaches attempt to learn fusion using approaches such as the use of the gating network and the attention-based fusion module.

Although these advances have been achieved, the area of multimodal action recognition in cricket using state-of-the-art pretrained models still lacks systematic investigation. The existing benchmark datasets fail to include the specific characteristics of cricket in the respective broadcasts, and state-of-the-art multimodal approaches tend to be computationally expensive and unadjustable for domain-specific problems. Moreover, comparisons involving various fusion techniques on cricket datasets remain uninvestigated.

The above-mentioned gaps therefore motivated this project to explore cricket video action recognition using a custom-created dataset. Given this, the study explores practical and scalable audio-visual fusion for cricket action recognition by making use of pre-trained visual models from ResNet18, R3D-18, and C3D, along with the extraction of several audio features such as YAMNet, VGGish, CLAP, and MFCC through an extensive examination of early, late, gated, and attention-based fusion strategies.

1.4 Problem Definition

In cricket action recognition, it is necessary to classify a number of video clips into semantically meaningful categories based on complex and noisy multimedia inputs. However, the problem of cricket action recognition proves challenging because of the similarity of different actions, consistency of cameras, background noises, and audio-visual misalignments.In this project, the task is to automatically classify cricket video clips into four categories: Four Runs, Six Runs, Wicket, and Others based upon the multimodal features of audio and video inputs. The aim of the project is to come up with a multimodal approach that robustly learns complementary features in both audio and video modalities in an effort to improve the accuracy of the classification task.

1.5 Objectives of the project

- **Audio Feature Extraction:** For extracting discriminative audio features from cricket video sequences employing a variety of methods such as YAMNet, VGGish, CLAP, and MFCC based on semantic and acoustic phenomena of audience reactions and sounds of bat and ball.

- **Visual Feature Extraction:** To extract high-level spatial and spatiotemporal features of the cricket video using pre-trained deep models of ResNet18, R3D18, and C3D to model the action and motion of the player.

- **Multimodal Feature Fusion:** The proposed work aims to use various audio-visual fusion techniques like early fusion, late fusion, gated fusion, and attention-based fusion to unify the audio and visual features in order to effectively analyze the effectiveness of this approach in recognizing actions in the game of cricket.

- **Model Design and Training:** Developing customized multimodal classification models to learn the combined audio visual features for the prediction of Cricket events like Four Runs, Six Runs, Wicket, and Others.

- **Performance Evaluation:** To test proposed multimodal approaches using conventional metrics for classification problems such as Accuracy, Precision, Recall, and F1-Score, and compare performance of multimodal approaches against unimodal baselines.

Chapter 2

SOFTWARE REQUIREMENT SPECIFICATION

The SRS described here intends to formulate a multimodal action recognition methodology for cricket videos with a classification setup for four events: Four Runs, Six Runs, Wicket, and Others for a dataset designed and developed by me.The proposed system intends to make use of visual feature extraction by models like ResNet18, R3D-18, and C3D and audio feature extraction by models like YAMNet, VGGish, CLAP, and MFCC. Various combinations of early fusion, late fusion, gated fusion, and attention-based fusion methods are also intended to be used for better performance. The proposal intends to use Python as the programming language and intends to make use of PyTorch, TensorFlow/Keras, and scikit-learn libraries for the making of this system.Precision and recall values along with Confusion Matrix are also used for evaluation.In this SRS document, all functional and non-functional requirements have been described for proper system development and subsequent improvements.

2.1 Overview of SRS

This SRS provides the overall description of the cricket action recognition system based on audio-visual data fusion. The system should classify cricket video clips into the following categories: Four Runs, Six Runs, Wicket, and Others correctly, by creating a dataset from real broadcast videos. This system would be utilizing pre-trained deep learning models for feature extraction like ResNet-18, R3D-18, and C3D for visual features, and YAMNet, VGGish, CLAP, and MFCC for audio features. Multiple strategies for fusion such as early fusion, late fusion, gated fusion, and attention-based fusion will be used to combine audio and visual information for a robust classification. The SRS functional and non-functional requirements are gathered, along with the design architecture, flow of data, and the methodology of evaluation. In addition, it ensures that the system is efficient, modular, and scalable, along with clear performance reporting and interpretability mechanisms. This document serves as a guide for developers, testers, and stakeholders to understand the scope and functionality of this system, along with its technical specifications.

2.2 Requirement Specifications

The functional requirements specify the functionalities that must be performed by the system for it to achieve its tasks. The project requirements under this criterion include loading and preprocessing video clips of crickets, the extraction of high-level visual features with the help of pre-trained models (ResNet18, R3D-18, C3D), the extraction of features for audio with the help of YAMNet, VGGish, CLAP, and MFCC, applying normalization and dimensionality reduction as needed, implementing early fusion, late fusion, gated fusion, and attention-based fusion, designing custom multimodal classifiers for the prediction of cricket events (Four Runs, Six Runs, Wicket, and Others), as well as performing stratified k-fold cross-validation for system evaluation, obtaining metrics for the system accuracy, precision, recall, F1-score, and generating the confusion matrix. The system shall be able to perform all of the above functionalities accurately, providing a valid cricket action recognition.

2.2.1 Functional Requirements

- The system shall load and preprocess the fused audio-visual features extracted from cricket video clips along with their corresponding labels for four action categories: Four Runs, Six Runs, Wicket, and Others.
- The system shall perform the training of multi-modal classifications with various fusion strategies.
- early, late, gated, and attention-based, respectively, on the fused features. The system should be able to support stratified k-fold cross-validation for balanced evaluation. across all action classes.
- The system shall evaluate model performance by generating metrics including accuracy, precision, recall, F1-score, and confusion matrices.

2.2.2 Use Case Description

Use Case Name: Cricket Audio-Visual Action Classification

Actors

- User (Researcher or Developer)
- System (Cricket Audio-Visual Recognition Ensemble System)

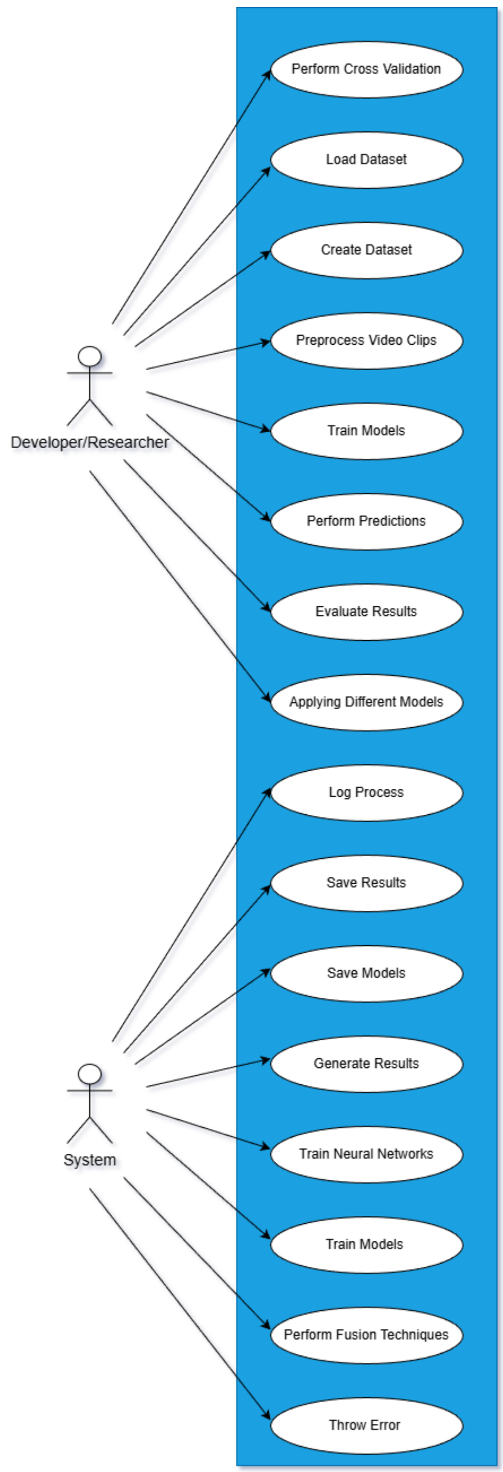

Figure 2.1: Use Case Diagram.

Preconditions

- The user provides fused features extracted from cricket audio models (YAMNet, VGGish, CLAP, MFCC) and video models (ResNet18, R3D-18, C3D).
- The dataset is preprocessed and labeled for training and testing.
- The system has access to saved feature files and model configurations.

Flow of Events

Basic Flow:

1. The user initiates the recognition process by loading the cricket dataset.
2. The system loads fused audio-visual features along with
3. The system performs stratified k-fold cross-validation.
4. Multimodal classification models are trained using different fusion strategies.
5. Predictions are generated on validation and test splits.
6. Ensemble or attention-based fusion combines model outputs for final predictions.
7. The system computes performance metrics including accuracy, precision, recall, F1-score, and confusion matrix.
8. Results, logs, and visualizations are saved for further analysis.

Exceptions

- **E1:** Input feature files are missing or corrupted; the system raises an error and terminates execution.
- **E2:** CPU or GPU resources exceed available capacity; the system prompts the user to reduce batch size or feature dimensionality.

Priority and Frequency

- **Priority:** High
- **Frequency of Use:** Frequent — during training, validation, and testing phases.

Special Requirements

- Minimum 16 GB RAM and an NVIDIA GPU (e.g., A100) for efficient training.
- Python environment with PyTorch, NumPy, scikit-learn, and Matplotlib libraries.
- Compatibility with custom cricket datasets and stored fused feature formats.

2.2.3 Nonfunctional Requirements

- The system shall complete each training epoch within a reasonable time (e.g., $\leq$15 minutes on GPU and $\leq$30 minutes on CPU).
- The system shall maintain an overall classification accuracy above 85% across all four cricket action categories using cross-validation.
- The system shall manage memory efficiently, limiting usage to approximately 16 GB during preprocessing and training.
- The system shall store logs, accuracy/loss plots, confusion matrices, and model checkpoints in a structured directory to ensure reproducibility.
- The system shall be modular and scalable, allowing the integration of new fusion strategies or additional models in the future.

2.2.4 Software and Hardware Requirement Specification

The software requirements for the proposed cricket audio-visual action classification system include Python 3.8 or higher as the primary programming language. PyTorch is used as the core deep learning framework for training neural networks and implementing gated and attention-based fusion models, while Scikit-learn is utilized for preprocessing tasks such as normalization, learning strategy. NumPy and Pandas are used for numerical computation and feature handling, and Matplotlib and Seaborn are used for visualization of accuracy plots, loss curves, and confusion matrices.

Audio features are extracted using pretrained and handcrafted models such as YAMNet, VGGish, CLAP, and MFCC, while visual features are obtained from deep convolutional and spatiotemporal networks including ResNet18, R3D-18, and C3D. Jupyter Notebook or Visual Studio Code is recommended for development and experimentation. Git is used for version control and collaborative management of source code and experiments.

From a hardware perspective, the system requires a multi-core CPU to support data loading. For accelerated training of deep neural networks, an NVIDIA GPU such as RTX 3060 (8 GB VRAM) or higher is recommended. The system should provide at least 500 GB of SSD storage to accommodate the cricket dataset, extracted audio-visual features, trained model checkpoints, and experimental results. Stable internet connectivity is required for downloading pretrained models and external dependencies.

2.2.5 Nonfunctional Requirements

- **NFR1: Performance** The system shall achieve a minimum classification accuracy of at least 85% across all defined cricket action categories using multimodal fusion strategies, including early, late, gated, and attention-based fusion.

- **NFR2: Scalability** The system will scale well to incorporate other classes of cricket action. audio-visual feature extractors, and extended datasets without significant degradation in classification accuracy and/or GPU memory consumption.

- **NFR3: Robustness** inputs influenced by crowd noise or commentary and video inputs influenced by motion blur OR lighting variations, by using fused multimodal representations.

- **NFR4: Reliability** The system will prove its performance capabilities on the stratified k-fold datasets. cross validation, with accuracy and F1 measure allowed to fluctuate ±2% validation folds.

Chapter 3

PROPOSED SYSTEM

This chapter discusses in detail the design and development framework of the proposed audiovisual. actions recognition system. With both audio and visual channels used in the system recognizes actions extracted from videos. The following subsections describe in detail of the system, its target users, benefits, uses, and coverage.

3.1 Description of Proposed System.

The proposed system is a multimodal deep learning architecture for action recognition in Cricket dataset, using audio visual cues for Action Recognition. It uses pretrained models of feature extraction, fusing modalities attention mechanism, and utilizes a variety of neural network models for robust action recognition. The system These modules include preprocessing, feature extraction, recognition, and evaluation.

3.2 System Modules

3.2.1 Data Processing Module

This module is used to process the custom cricket audio-visual data for training and testing purposes. The audio-visual data is a collection of video files containing various actions performed in a cricket match (e.g., batsman shots, boundaries, wickets, and more). The data is accompanied by audio files depicting these actions and is used in this module to be labeled and then used for supervised learning. The class labels are converted into numerical variables to facilitate supervised learning processes. The audio and video features are then randomly shuffled using a predefined random state to ensure reproducibility for each experiment. The stratified split is also used to ensure equality is maintained between classes when splitting the data into training and testing sets. The distribution of samples across each class is then graphed using bar plots to analyze potential imbalances.

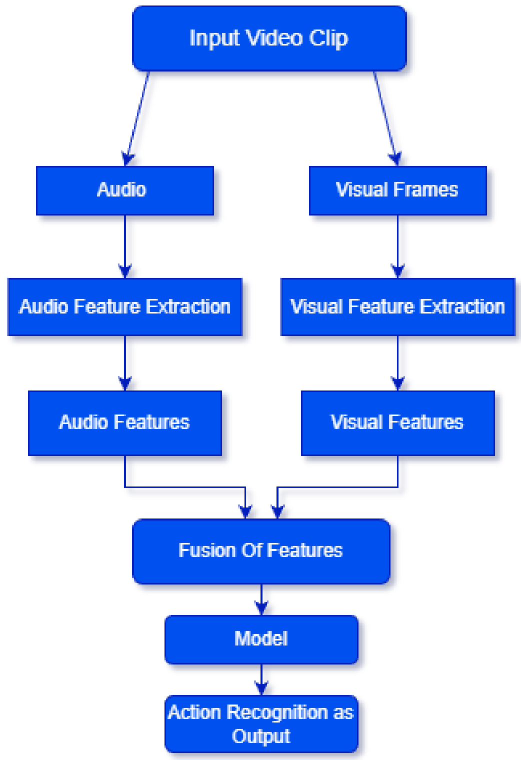

Figure 3.1: Block Diagram.

3.2.2 Feature Extraction Module

The feature extraction module obtains the high-level representations of the cricket videos from the visual and auditory perspectives. The visual features are extracted from the cricket videos by means of pre-trained deep learning models ResNet-18, R3D-18, and C3D, which focus on the spatial and spatio-temporal patterns of cricket motion. The video images are uniformly sampled, resized to the standard input size, transformed into the RGB color space, and normalized. The audio features of the cricket videos are extracted from the audio streams by means of YAMNet, VGGish, CLAP, and MFCC, which focus on the sound of the crowd, the sound of the bat and the ball interaction, as well as the commentary.

3.2.3 Classification and Fusion Module

This module performs the multimodal action classification by fusing audio and visual features under multiple fusion strategies, namely early fusion, late fusion, gated fusion, and attention-based fusion. Early Fusion: Audio and video features are concatenated before classification. Late Fusion: Prediction scores from independently trained unimodal classifiers are combined. Gated Fusion: Modality importance is learned dynamically through trainable gating mechanisms. Attention-based Fusion: It allows selectively emphasizing the discriminative temporal and modality-specific features. Classification Models: For these, the code provides the implementation of a custom multimodal neural network model, which will be optimized using the Adam method with appropriate loss functions, and early stopping to avoid overfitting.

3.2.4 Evaluation and Visualization Module

The evaluation tool examines system performance by utilizing a set of traditional classification evaluation metrics such as accuracy rate, precision rate, recall rate, and F1-score values. The categorical values considered in classification analysis relate to a set of actions in a cricket match called cricket actions. The performance evaluation of different multimodal setups using feature extractors is carried out, and system performance is compared to find optimal solutions/strategies. Results of evaluations carried out in every process stage of the system are saved in a proper manner.

3.2.5 System Overview

Figure 3.1 presents a general block diagram of the proposed cricket audio-visual action recognition system, showing interactions among preprocessing, feature extraction, fusion, classification, and evaluation modules.

3.3 Description of Target Users

It forms a system that especially enables its users in the development, evaluation, and deployment of audio-visual action recognition systems with a key focus on sports analytics. This shall be an efficient guide for researchers, data scientists, and developers operating under the computer vision, multimedia analysis, and multimodal machine learning domains. It particularly finds relevance in applications related to cricket analytics, including automated highlight generation, performance analysis, and event detection from match videos. Secondary users include academic institutions doing research into multimodal learning and industry teams developing AI-driven sports video analysis platforms. The modular and extensible nature of the system also makes it suitable for experimentation with new fusion strategies and feature extraction models.

3.3.1 Design Principles

- **Modularity:**The system is divided into independent modules on data preprocessing, feature extraction, multimodal fusion, classification, and evaluation. This structure easily enables replacing or extending parts of it-for instance, adding other audio or video models or testing new fusion techniques.
- **Robustness:**The proposed model utilizes pre-trained audio and visual models and various fusion strategies such as early, late, gated, and attention-based fusion to handle variation in video quality, background noise, and recording conditions of real-world cricket footage.
- **Reproducibility:**Fixed random seeds, standardized preprocessing pipelines, and uniform evaluation protocols stratified k-fold cross-validation are used to guarantee repeatable and reliable experimental results across different runs and configurations.

3.4 Advantages/Applications of Proposed System

T his sub-section summarizes some of the main benefits and application areas of the cricket audio-visual action recognition proposed system; it further emphasizes the multimodal architecture and real-world effectiveness in sports data analysis.

3.4.1 Advantages

- **Audio-Visual Multimodal Integration:** It fuses complementary audio and visual cues like bat–ball impact sounds with the motion of players and it achieves better recognition accuracy compared to vision-only approaches for events recognition in cricket for Four Runs, Six Runs, Wicket, Others.
- **Robust Performance Across Conditions:**By utilizing multiple models pretrained and early, late, gated, and attention-based fusion schemes, the system is pretty consistent when there is variation in camera angles, crowd noise, and quality of videos.
- **Modular and Scalable Architecture:**The modular pipeline allows seamless integration of new models, fusion strategies, or new cricket action classes. Moreover, it may work for other sports or extended datasets.

- **Comprehensive Evaluation Framework:**Stratified k-fold cross-validation with detailed performance metrics ensures the evaluation is consistent, reproducible, and unbiased for all model combinations.

3.4.2 Applications

- **Cricket Match Analytics:**This will automatically detect the key match events, such as boundaries and wickets, from which a variety of performance analyses, statistics generation, and summarization are possible.
- **Automated Highlight Generation:**This identifies the high-impact moments (e.g., sixes and wickets) from a long video of cricket and allows broadcasters and digital platforms to prepare fast and efficient highlights.
- **Sports Video Indexing and Retrieval:**It provides a structured tagging and indexing of videos in cricket, improving the search, retrieval, and content management for sports media archives.
- **AI Research and Education:**It would be a fundamental framework for benchmarking custom datasets for experimentation of multi-modal fusion strategies for sports action recognition.

3.5 Scope

3.5.1 Boundaries:

- The system processes pre-recorded cricket video clips incorporating visual frames and audio signals extracted from a cricket dataset designed by me.
- Extraction of visual features is restricted to deep-trained models ResNet18, R3D-18, and C3D, whereas the extraction of audio features is performed via YAMNet, VGGish, CLAP, or MFCC.
- Action recognition is confined to four predefined classes regarding cricket actions, namely, *Four Runs, Six Runs, Wicket, Others*
- The system is based on offline processing, where either feature extractions are performed beforehand or feature files are loaded for training and testing purposes.

- Multimodal fusion is performed using early fusion, late fusion, gated fusion, and attention-based fusion strategies within deep learning and ensemble-based classification models.

3.5.2 Future Directions:

- Extension of the system to support real-time or near real-time cricket video analysis, enabling live match event detection.
- Deployment on edge devices or resource-constrained platforms, such as sports analytics hardware or mobile systems.
- Expansion of the dataset to include additional cricket events (e.g., no-balls, wides, catches) and other sports.
- Integration with live streaming platforms for automated highlight generation and real-time sports analytics.

Chapter 4

SYSTEM DESIGN

The system design helps to outline the interaction among the different modules of the proposed cricket audio-visual action recognition system, assisting in the correct classification of significant cricket events. The design of the cricket action recognition system is based on the multimodal learning approach, which combines the audio and video inputs of the cricket scenes. The following section illustrates the different aspects of the cricket action recognition system.

4.1 System Architecture

4.1.1 Data Loading and Preprocessing Module

- **Input:**The proposed system is implemented on an internally generated cricket video dataset that contains four action categories of pre-recorded videos: Four Runs, Six Runs, Wicket, & Others. Metadata and labels are loaded from structured files, and the dataset is split into a training, validation, and testing set. Data shuffling is carried out by fixing a random number or seed so that it is reproducible.

- **Visual Preprocessing:**The video clips are decoded and sampled for extraction of representative frames/short clips based on the visual backbone network. The frames are resized to have a constant size (for example, 224×224 pixels), transformed from BGR to RGB color channels, and normalized as required by the input of the respective pretrained models.

- **Audio Preprocessing:**Audio segments are derived using typical video processing libraries from cricket videos and resampled at a constant sample rate. Audio is either converted to MFCCs, log mel-spectrograms, and/or raw waveform representations based on feature extraction techniques used in extractors.

- **Label Encoding:**Action labels are encoded into numerical form to support supervised learning and evaluation across all models.

4.1.2 Feature Extraction Module

- **Visual Feature Extraction:**Visual features are extracted using pretrained deep models: ResNet18 for frame-level spatial features,R3D-18 for spatio-temporal video representations,C3D for short-term motion and temporal pattern learning.Extracted features are aggregated (mean or temporal pooling) to form compact video-level embeddings.

- **Audio Feature Extraction:**Audio representations are extracted using multiple pretrained and handcrafted approaches:YAMNet for general audio event embeddings,VGGish for deep audio feature representations,CLAP for cross-modal audio embeddings,MFCC for low-level spectral features.Temporal pooling is applied to obtain fixed-length audio vectors.

4.1.3 Multimodal Fusion and Training Configuration

- **Fusion Strategies:**The system evaluates multiple multimodal fusion mechanisms: Early Fusion forFeature-level concatenation of audio and visual embeddings.Late Fusion for Decision-level combination of independently trained audio and video models.Gated Fusion for Learnable gating mechanisms that dynamically weigh each modality.Attention-Based Fusion for Attention layers that adaptively emphasize the most informative modality.

- **Training Configuration:**Models are trained using the Adam optimizer with appropriate loss functions and early stopping to prevent overfitting. Stratified k-fold cross-validation is employed to ensure stable and unbiased evaluation.

4.1.4 Dataset Creation and Annotation

A custom cricket action dataset was created specifically for this project to address the lack of publicly available datasets containing fine-grained cricket events with synchronized audio and visual information. Figure 4.1 shows sample image of Custom Dataset. Full match videos for the sport of cricket were gathered from publicly available sources YouTube sources and downloaded on a local computer for processing. For every video, key events related to cricket plays such as Four Runs and Six were manually extracted. These events were used to calculate scores that Runs, Wicket, and Others. The events annotated in the excel spreadsheet included detailed information involving videoId, event label, start time, end time, and clip. duration. A spreadsheet with annotations was further translated into a CSV file to

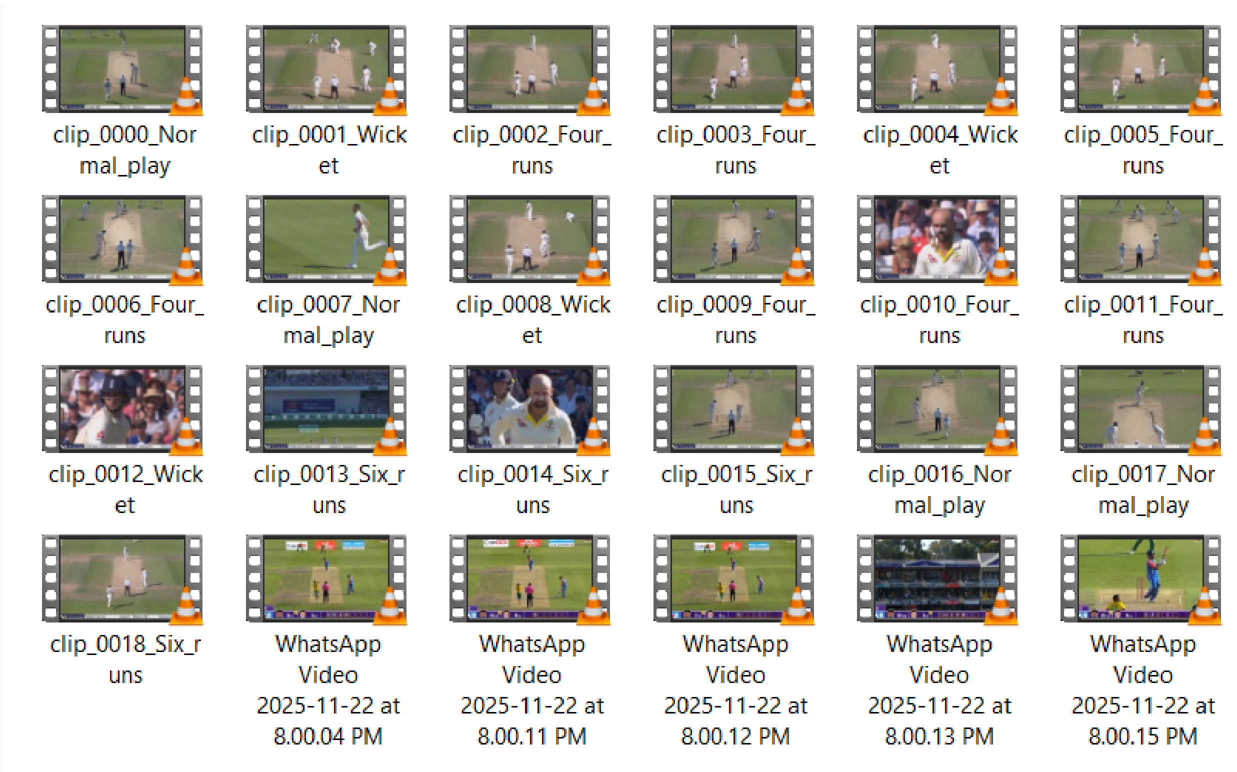

Figure 4.1: Sample image of Dataset

support automated processing. This custom script was created using Python programming to parse and clip videos based upon annotated start and end points of the videos. Each clip extracted was automatically saved in a corresponding class-specific directory (Four runs, Six runs, Wicket, Others), thus obtaining a well-organized and label-consistent dataset. Such automatic clipping The pipeline eliminates human error, preserves the accuracy of time-related information, and facilitates scalability of the dataset expansion.Each action category folder (Four runs, Six runs, Wicket, Others) has 350 video clips, thus ensuring that the dataset is balanced for all classes. To enable effective trai evaluation, the dataset was divided into training, validation, and test datasets in the ratio of 0.8:0.1. : 0.1 ratio

Dataset Limitations: There are many limitations to the custom cricket dataset. The dataset contains videos taken from YouTube, which can cause uniform resolutions, illuminations, views, and noises to become uncontrolled variables. There can also be some inaccuracies when done manually, especially when it comes to overlapping occurrences. The dataset only contains four actions with only a few instances of each. The videos are edited, not streamed live.

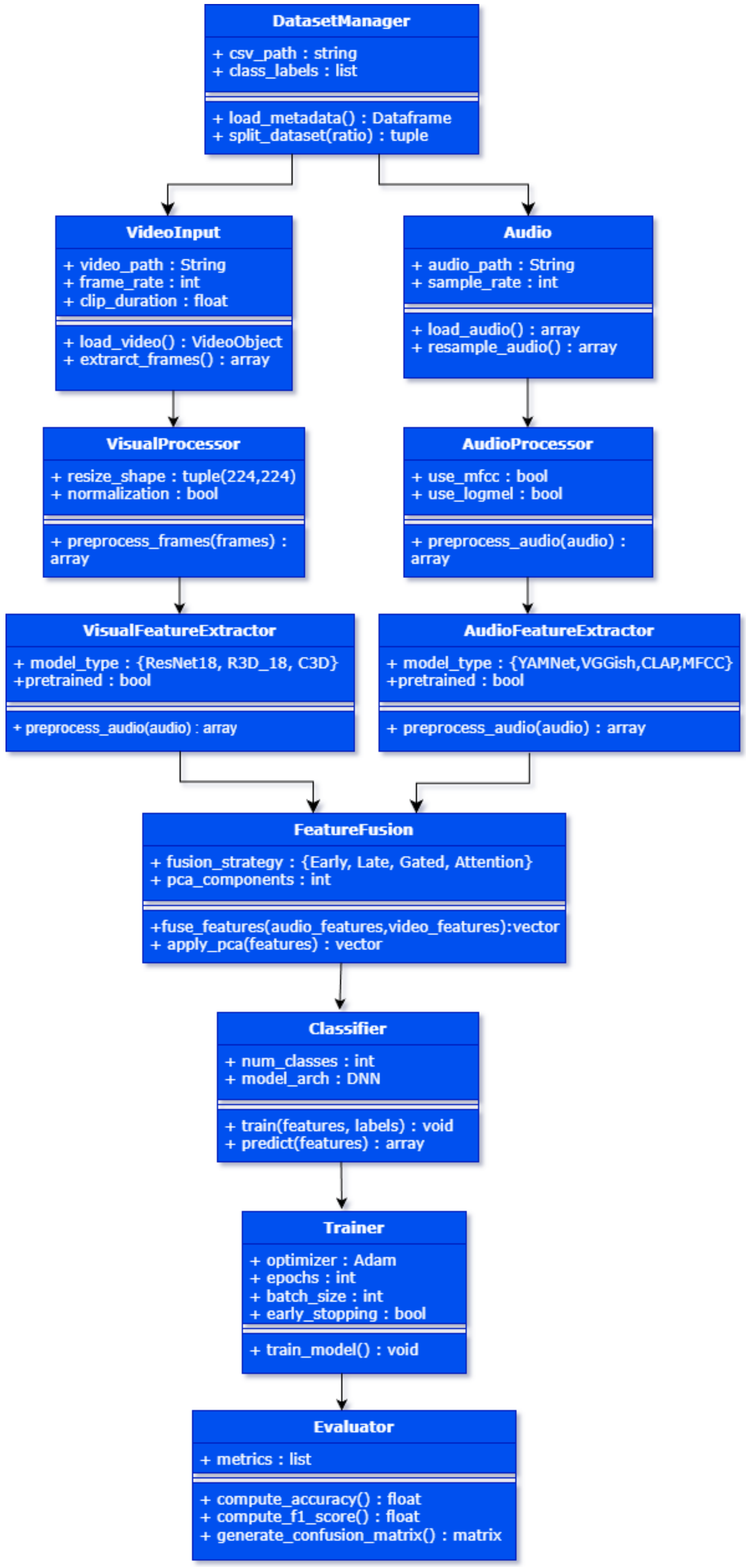

Figure 4.2: Class Diagram of System

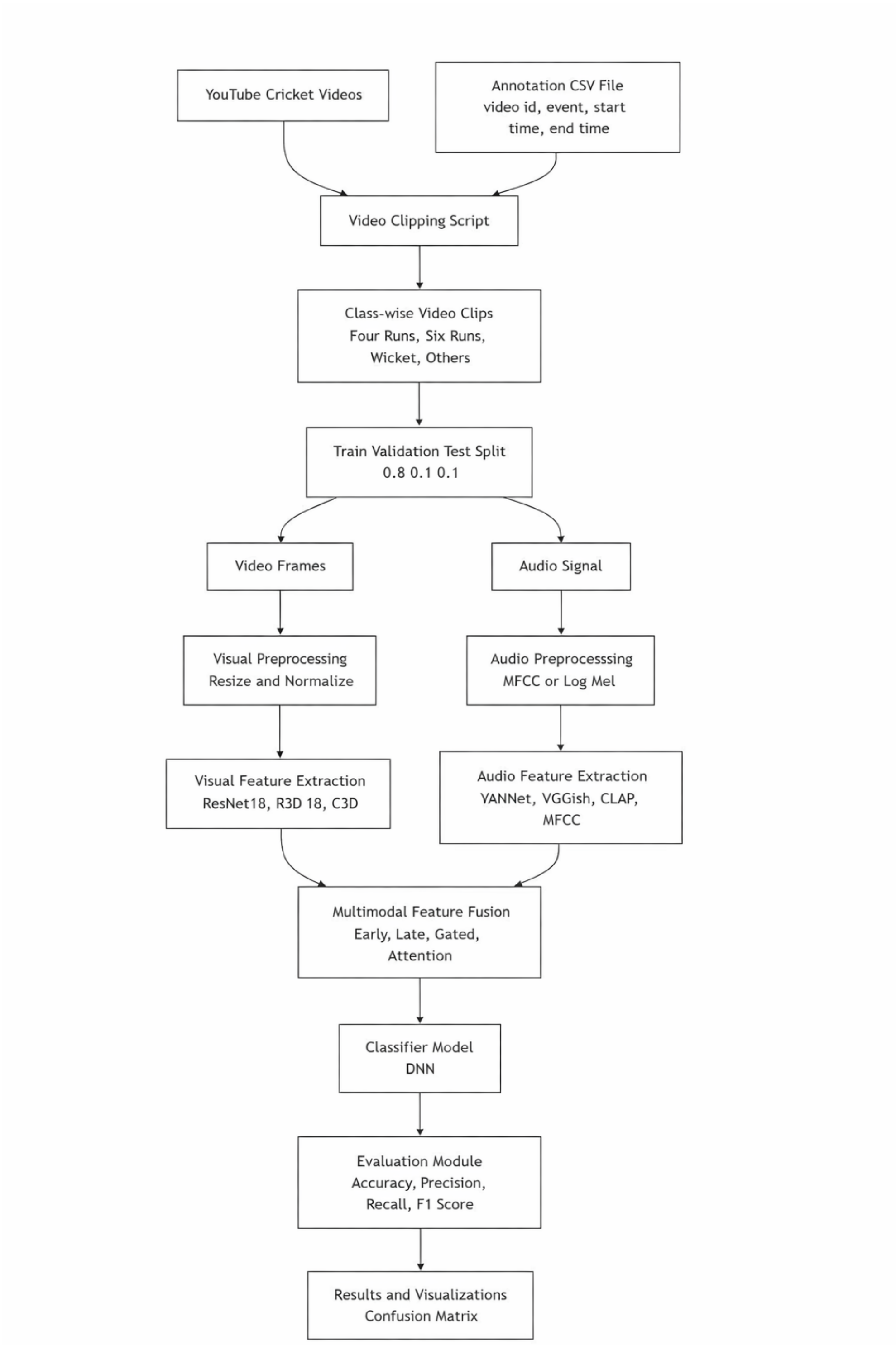

Figure 4.3: Dataflow Diagram of System

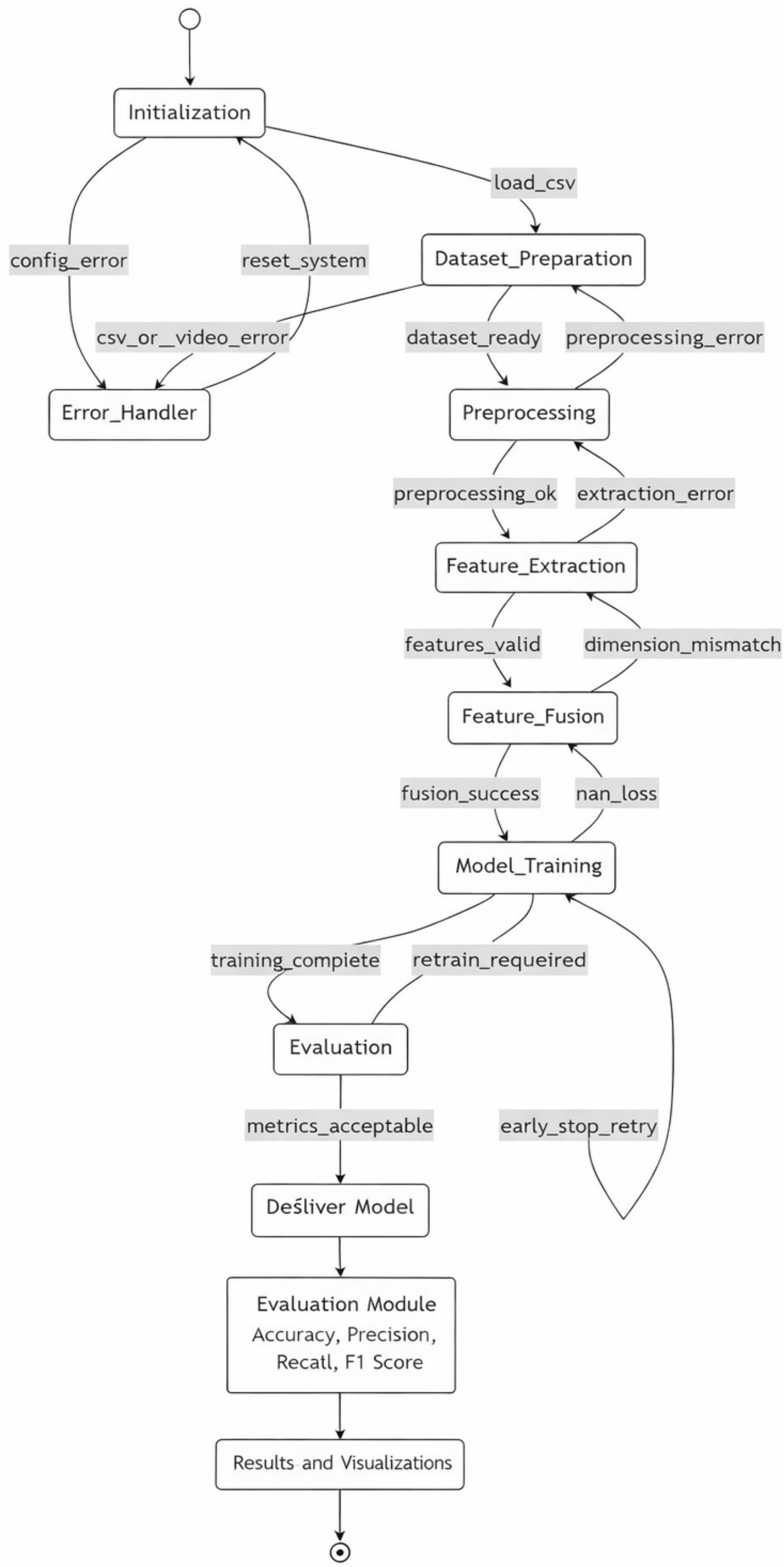

Figure 4.4: State Transition Diagram of System

Chapter 5

IMPLEMENTATION

Based on the metrics measured on the test set, the ResNet18 + YAMNet with attention-based fusion performs best for all combinations of model and fusion type in achieving an accuracy of about 86%, outperforming other methods consistently. Unlike the other methods such as MFCC + C3D or R3D-18 + MFCC, ResNet18 + YAMNet with attention-based fusion performs relatively equally on different classes for Four runs, Six runs, Wicket, and Others in terms of precision, recall, and F1-score. The confusion matrix shows strong diagonal dominance, indicating reliable predictions, with limited misclassification between visually similar actions such as Four runs and Six runs, and no severe confusion where boundary events are misclassified as Others. The attention-based fusion mechanism effectively enhances performance by selectively weighting audio and visual cues, which is particularly important in cricket action recognition where audio signals help distinguish bat impact events while visual motion provides contextual confirmation. Furthermore, YAMNet outperforms MFCC for this task by capturing richer, pretrained semantic audio representations, leading to improved discrimination of fine-grained sports sounds and overall robustness of the system.

5.1 Proposed Methodology

The proposed methodology focuses on audio-visual action recognition using a unified and efficient deep learning pipeline built on pretrained feature extractors and a DNN-based attention fusion mechanism. In contrast to using multiple heterogeneous models based on different architectures, the approach employs a unified stable design for the model so that fairness in comparisons is ensured for different methods of fusion.

The features are extracted using ResNet-18 for visualization features and YAMNet for audio features, as it is a pre-trained model for audio event classification. The features are combined using various methods, of which attention-based fusion is found to perform best. The combined features are then classified using multilayer perceptron (DNN) with Adam optimizer and early stopping.Fig 5.1 shows architecture of proposed system.

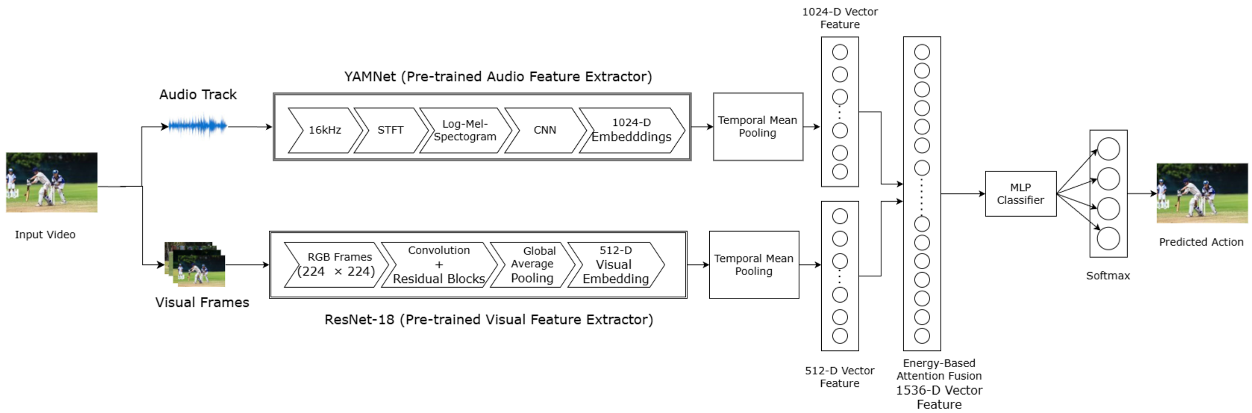

Figure 5.1: Proposed Methodology

5.2 Description of Modules

This system performs audio–visual action recognition from video data, where each well-defined module works together in building up the pipeline, starting with data preparation and feature extraction, through attention-based fusion, to classification and performance evaluation.

5.2.1 Dataset and Preprocessing Module

This module handles dataset organization and preprocessing. Video clips are divided into training, validation, and test sets using an 80:10:10 split while maintaining class balance. Each video is decomposed into its audio track and visual frames. Class labels are encoded numerically for supervised learning. The dataset preparation ensures that both audio and visual modalities are synchronized and suitable for multimodal learning.

5.2.2 Feature Extraction Module

The feature extraction module independently processes visual and audio data using pre-trained deep learning models.

Figure 5.2 illustrates Visual Features extracted in RGB frames resized to 224×224 are passed through a pretrained ResNet-18 network. Global Average Pooling is applied to obtain 512-dimensional visual embeddings, which are then temporally mean-pooled across frames to generate a single visual representation per video.

Figure 5.3 shows how the audio track is resampled to 16 kHz and processed using YAMNet. The audio is transformed into log-mel spectrograms internally, and the network produces 1024-dimensional audio embeddings. Temporal mean pooling is applied to aggregate

Figure 5.2: Visual Frames

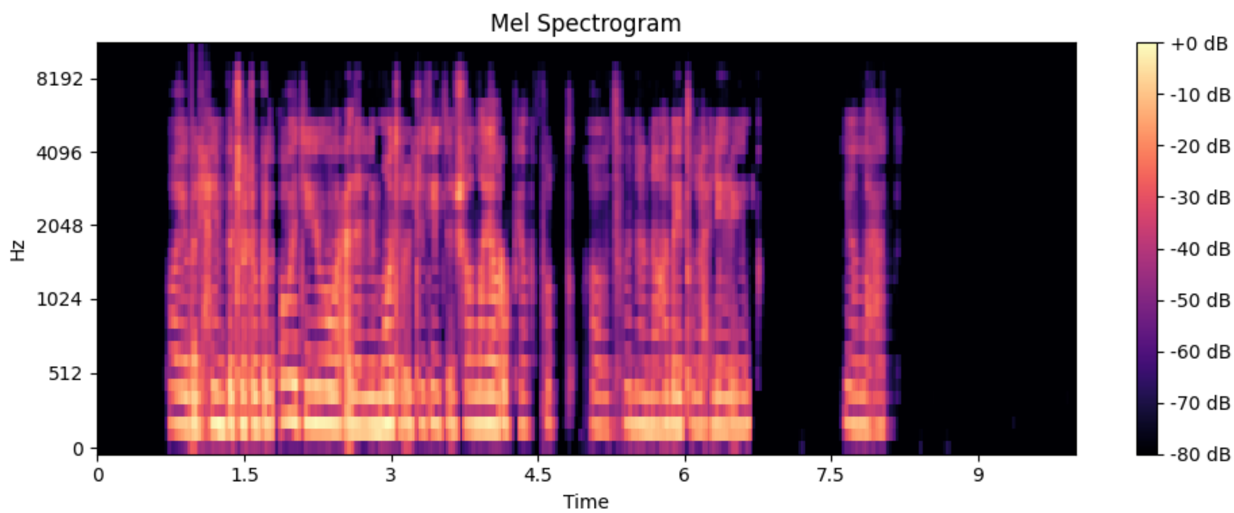

Figure 5.3: Log-Mel Spectogram

embeddings across time, resulting in a fixed-length audio feature vector. This design leverages pretrained semantic representations while avoiding manual feature engineering.

5.2.3 Attention-Based Fusion and Classification Module

The extracted audio (1024-D) and visual (512-D) features are combined using an attention-based fusion mechanism. The attention module dynamically assigns weights to each modality based on their relevance, enabling the model to emphasize audio cues (e.g., bat impact sounds) or visual context (e.g., player motion) as required. Fig. 5.4 and 5.5 shows how MLP classifier predicting action with confidence level. The weighted features are fused into a 1536-dimensional representation, which is passed to a multilayer perceptron (DNN) classifier. The classifier consists of fully connected layers with ReLU activation and a final

Figure 5.4: Action Recognition Sample 1

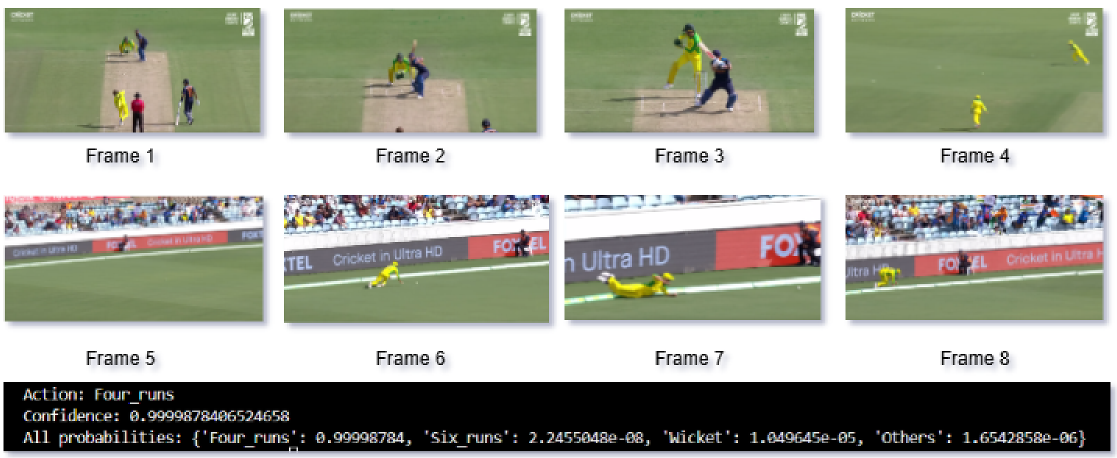

Figure 5.5: Action Recognition Sample 2

softmax layer to perform four-class action recognition. Model training is carried out using the Adam optimizer, with early stopping employed to prevent overfitting.

5.2.4 Evaluation and Visualization Module

This module assesses the performance of the proposed system based on standardized measures such as accuracy, precision, recall, and F1 measures. In this case, classification reports are carried out to analyze the performance of the system with respect to each class. Confusion matrices are utilized to demonstrate correct classification and classification errors, which give insights on inter-class confusion, especially between similar actions performed by cricketers based on visual and audio information.

Chapter 6

TESTING

In this work, a systematic testing process has been carried out on the proposed audio-visual action recognition system using the custom dataset of cricket actions, so as to ensure correctness, reliability, and robustness. This system focuses on four action classes: Four runs, Six runs, Wicket, and Others, involving a ResNet18-based visual feature extractor, YAMNet-based audio feature extractor, and attention-based fusion DNN classifier. Testing of all the critical stages of the pipeline includes dataset loading, video clipping, feature extraction, attention fusion, model training, inference, and performance evaluation.

The testing process actually validated that this end-to-end pipeline is working as it should, taking a raw input video to the very final step of predicting actions with confidence scores. Most test cases have passed without any issues and brought in feature stability with respect to extraction and classification. Some minor issues, such as missing audio tracks or corrupted video clips, were found during testing and handled with validation checks. Passed and failed test cases also provide insights into system robustness and suggest areas where careful data preprocessing is needed.

6.1 Passed Test Cases

The test case depicted in the subsequent table 6.1 was successfully used to test the functionality of the designed system. The processing of datasets was carried out successfully to produce labeled videos. The visual features successfully extracted through ResNet18 generated 512-dimensional embeddings, while audio features extracted through YAMNet generated 1024-dimensional embeddings. The attention mechanism fused the features effectively to get a unified representation, and the DNN classifier trained successfully. The metrics of evaluation, including metrics of accuracy, class reports, and confusion matrices, have been produced successfully.

Table 6.1: Passed Test Case Validation Results

Test Case ID	Test Scenario	Expected Output	Result
TC_01	Load dataset CSV file	Valid metadata loaded with 4 classes	Pass
TC_02	Video clipping using timestamps	Class-wise video clips generated	Pass
TC_03	Extract visual frames	Frames resized to 224×224	Pass
TC_04	Extract audio track	Audio sampled at 16 kHz	Pass
TC_05	Visual feature extraction (ResNet18)	512-D feature vector	Pass
TC_06	Audio feature extraction (YAMNet)	1024-D feature vector	Pass
TC_07	Attention-based fusion	Weighted fusion of audio & visual probabilities	Pass
TC_08	Train DNN classifier	Stable convergence	Pass
TC_09	Model inference on test video	Correct action prediction with confidence	Pass
TC_10	Performance evaluation	Confusion matrix & classification report	Pass

6.2 Failed Test Cases

Table shows 6.2 A limited number of test cases failed due to data-related issues rather than model design flaws. Videos with missing or silent audio tracks caused YAMNet feature extraction to fail, leading to incomplete fusion inputs. Corrupted or improperly encoded video files resulted in frame extraction errors. These failures highlight the importance of strict data validation and preprocessing when working with real-world multimedia datasets.

Table 6.2: Failed Test Case Validation Results

Test Case ID	Test Scenario	Expected Output	Result
TC_01	Audio feature extraction on silent video	Valid audio embedding	Fail
TC_02	Frame extraction from corrupted video	Valid frame sequence	Fail

Chapter 7

RESULTS & DISCUSSIONS

This chapter provides an in-depth analysis of the experiments conducted on the multimodal cricket action recognition system using combined audio-visual features. This analysis is performed on a self-generated dataset of cricket videos, which comprises Four Runs, Six Runs, Wicket, and Others classes. In our system, a Deep Neural Network (DNN) classifier is used, which is trained on the combined extracted features from several state-of-the-art audio & visual models. This analysis mainly emphasizes the use of multimodal fusion and how the DNN classifier is able to distinguish among the visually and aurally similar events in a cricket match.

7.1 Evaluation Metrics

The performance of the classification system developed using DNN can be objectively measured using the following parameters:

- **Accuracy:**Accuracy is calculated based on the overall correctness of the predictions made for each class
- **Precision:** References the number of correctly predicted action instances.
- **Recall:**It calculates the ability of a model to spot all actual instances of a specific action.
- **F1-Score:** Offers an equal weightage to the precision and recall measures. This is quite essential, as it gives class wise performance.

7.2 Confusion Matrix:

The confusion matrix gives a better understanding about how well the proposed system distinguishes among various activities related to cricket. It shows that the DNN works well on visually and acoustically distinctive events such as Six Runs and Wicket, while occasional Confusion between Four Runs and Others is noted to arise, mainly because of similarities in the crowd noise and visual motion patterns. In summary, the confusion matrix made it clear that the combination of these factors in the simulation audio and visual features simultaneously, and this has been shown to significantly reduce ambiguity when compared to single modal solutions. validating the efficacy of the proposed framework based on multimodal DNNs by measuring These evaluations have been carried out by performing stratified k-fold cross-validation tests.

Table 7.1: Unimodal Performance

Model	Visual Model	Audio Model	Visual Accuracy (%)	Audio Accuracy (%)
1	ResNet18	YAMNet	0.86	0.49
2	R3D_18	MFCC	0.74	0.60
3	ResNet18	VGGish	0.68	0.76
4	C3D	CLAP	0.76	0.47
5	C3D	MFCC	0.76	0.66

From Table 7.1, it is evident that there is always a trade-off between the performance of the visual and audio parts in the different model combinations. Model 1, ResNet18-Visual and YAMNet-Audio, displays the highest visual accuracy of 0.86 but the poorest audio accuracy of 0.49. Model 2, R3D_18-Visual and MFCC-Audio, reduces the level of visual accuracy to 0.74 but increases the level of audio accuracy to 0.60. Model 3, ResNet18-Visual and VGGish-Audio, reduces the level of visual accuracy to 0.68 but displays the best possible audio accuracy of 0.76. Model 4, C3D-Visual and CLAP-Audio, maintains the moderate level of visual accuracy of 0.76 but displays the poorest audio accuracy of 0.47. Model 5, C3D-Visual and MFCC-Audio, displays balanced accuracy of 0.76 and 0.66 in the visual and audio parts respectively. From these models, ResNet18 is best suited to the visual tasks, and the MFCC is best suited to the audio tasks. VGGish enhances the audio tasks but deteriorates the visual accuracy in Model 3. C3D provides moderate visual accuracy, while the audio accuracy is dependent on the model used.

Table 7.2: Performance of Models on Different Fusion Techniques with DNN based pipeline

Model	Visual Model	Audio Model	Early Fusion	Late Fusion	Gated Fusion	Attention Fusion
1	ResNet18	YAMNet	0.788	0.85	0.86	0.86
2	R3D-18	MFCC	0.79	0.74	0.77	0.77
3	ResNet18	VGGish	0.76	0.76	0.78	0.80
4	C3D	CLAP	0.837	0.873	0.86	0.80
5	C3D	MFCC	0.55	0.76	0.75	0.77

Table 7.2 summarizes the performance of the proposed audio-visual action recognition system using different combinations of visual and audio feature extractors under four fusion strategies: Early Fusion, Late Fusion, Gated Fusion, and Attention-based Fusion. The outcome of the comparison clearly shows that fusion strategies used influence the classification accuracy, and Late Fusion and Attention-based Fusion strategies performed better than Early Fusion on most of the models except Deep Neural Network based models.

Earlier stages of fusion report relatively poor results forAccuracy for most models on combination of models, especially for models R3D-18 + MFCC features and ResNet18 + VGGish features. These outcomes mainly arise because ofina Chufront of heterogeneous features between audio and visual features, disregarding the weights for these features. Difficulty is encountered in stages where semantics of features vary largely.

The late fusion has the highest or nearly highest accuracy in several combinations: YAMNet + ResNet18 (0.85) and C3D + CLAP (0.873). By allowing independent learning of audio and visual representations before combining prediction scores, late fusion preserves modality-specific discriminative power and reduces cross-modal interference.

Gated fusion demonstrates stable performance by dynamically controlling the contribution of audio and visual features. While it outperforms early fusion in most cases, it does not consistently surpass late fusion, indicating that gating mechanisms may be sensitive to training data size and class imbalance in the custom cricket dataset.

Although the C3D + CLAP model with late fusion achieved the highest overall accuracy (87.3%), the ResNet18 + YAMNet model with attention-based fusion demonstrated more balanced class-wise performance and superior robustness across all action categories. Considering accuracy, confusion patterns, and fusion effectiveness together, the attention-based audio–visual model was identified as the most reliable and practically applicable solution.

Overall, the ResNet18 + YAMNet model with attention-based fusion provides the best trade-off between accuracy, robustness, interpretability, and practical applicability, making it the most suitable architecture for audio-visual cricket action classification in this work.

7.2.1 Model 1 : Attention Fusion on YAMNet + ResNet18

Fig. 7.1 illustrates For Four runs, the model correctly classifies 62 samples, with minor confusion mainly with Wicket and Others, showing good discrimination of boundary shots. Six runs achieves 61 correct predictions with very limited misclassification, demonstrating reliable recognition supported by strong audio-visual cues. The Wicket class records 54 correct predictions, with some confusion with Four runs and Six runs, reflecting visual similarity between batting and wicket-taking actions. Others shows 58 correct classifications,

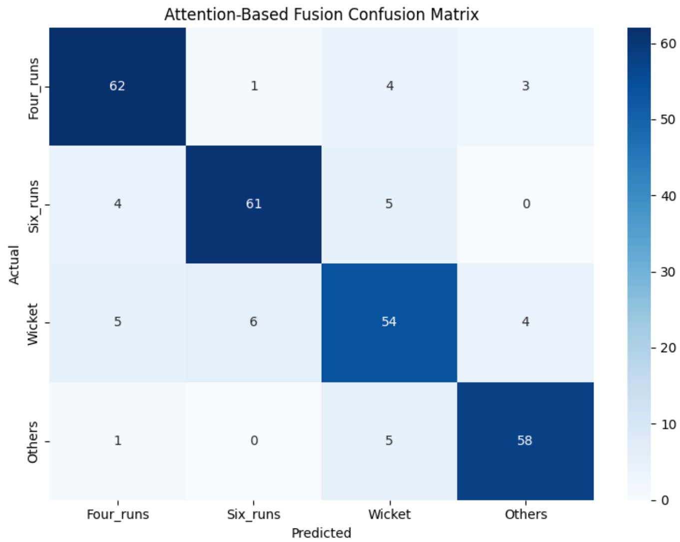

Figure 7.1: Confusion Matrix of Model 1 on Attention Fusion Mode

with minimal confusion, mainly with Wicket.Overall, the confusion matrix confirms that the attention-based fusion model effectively distinguishes between classes, with only limited overlap in visually or acoustically similar events.

From Table 7.3 The classification report provides a quantitative breakdown of the model's performance for each class using precision, recall, and F1-score. Six runs achieves the highest precision (0.90), indicating that when the model predicts a six, it is usually correct.Others has the maximum recall value of 0.91, indicating the model's capability to extract various non-boundary events effectively.Four runs maintains equal values for precision and recall, ensuring a stable F1-score of 0.87, implying effective detection. The values for Wicket class express lower values for precision (0.79) and recall (0.78), re-asserting confusion due to visual and audio resemblance of wicket class events to batting actions.Older and weighted average values are 0.86 for both values, implying equal performance of all classes without any inclination towards a particular class. The overall classification report asserts effective and balanced detection of various events by the attention fusion model.

Table 7.3: Classification Report of Model 1 on Attention-Based Fusion

Class	**Precision**	**Recall**	**F1-score**
Four_runs	0.86	0.89	0.87
Six_runs	0.90	0.87	0.88
Wicket	0.79	0.78	0.79
Others	0.89	0.91	0.90

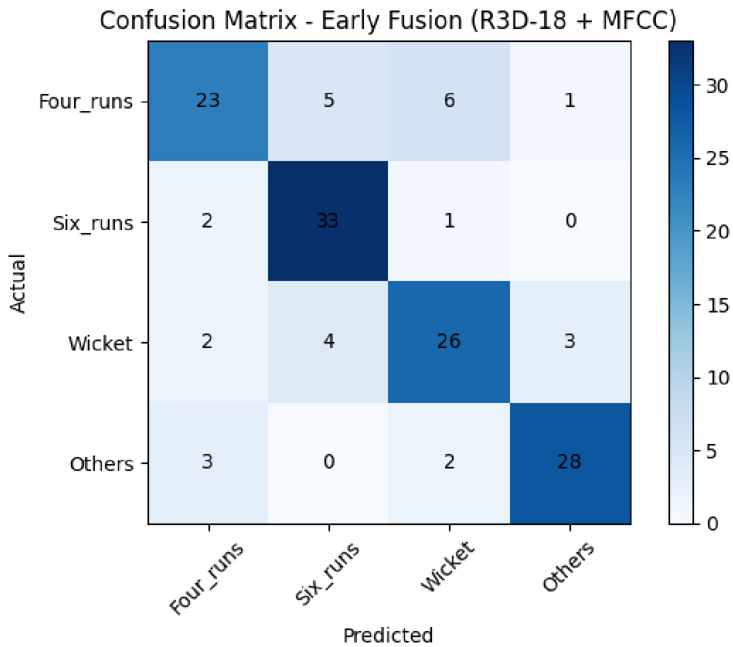

Figure 7.2: Confusion Matrix of Model 2 on Early Fusion Mode

7.2.2 Model 2 : Early Fusion on R3D_18 + MFCC

The above Figure 7.2 explains the confusion matrix and implies that the classifier is strongest on Six runs and Others, where there is prominent prediction accuracy, indicating good separability between various classes. The Six runs have maximum correct prediction, that is, 33 out of 36, and there is little overlap, which is mostly between Four runs, implying that there is some overlap between their motion features. The Four runs have maximum overlap, where they are mostly confused between Six runs and Wicket, which indicates that you are not aware that their visual and audio signals are not unique enough to distinguish between various run actions, which are of short duration. Wicket has moderate overlap, which indicates ambiguity between sudden motion visuals and various other classes, explaining sudden motion incidents. The above Figure 7.2 explains the confusion matrix and implies that the classifier is strongest on Six runs and Others, where there is prominent prediction accuracy, indicating good separability between various classes. The Six runs have maximum correct prediction, that is, 33 out of 36, and there is little overlap, which is mostly between Four runs, implying that there is some overlap between their motion features. The Four runs have maximum overlap, where they are mostly confused between Six runs and Wicket, which indicates that you are not aware that their visual and audio signals are not unique enough to distinguish between various run actions, which are of short duration. Wicket has moderate overlap, which indicates ambiguity between sudden motion visuals and various other classes, explaining sudden motion incidents.

Table 7.4: Classification Report of Model 2 on Early Fusion

Class	Precision	Recall	F1-score
Four_runs	0.77	0.66	0.71
Six_runs	0.79	0.92	0.85
Wicket	0.74	0.74	0.74
Others	0.88	0.85	0.86

7.2.3 Model 3 : Attention Fusion on ResNet18 + VGGish

The confusion matrix and classification report in Fig. 7.3 present the performance of Model 3, ResNet18 + VGGish, using attention-based fusion. The confusion matrix has strong diagonal dominance, meaning most video samples are correctly classified for all four action classes. The Four runs and Six runs classes have high correct prediction counts with low confusion, indicating effective discrimination of scoring-related events. The Others class also has good performance, with very few misclassifications, which indicates good generalization for non-specific actions. However, the Wicket class has noticeable confusion with Six runs and Four runs due to difficulties in distinguishing between visually and acoustically similar events in cricket. Overall, this matrix confirms attention fusion improves multimodal feature alignment but still struggles with fine-grained action boundaries.

From Table 7.5 above, The classification report provides further evidence to support these findings. Four runs have a high recall value of 0.91, which indicates that almost all actual examples are perfectly identified, although it has a low precision value because there are some false positives. Six runs have balanced precision and recall values (0.81), which indicates that there are stable and consistent outcomes. The Wicket class has very high precision (0.95) and low recall value (0.54), which indicates that there are stable and confident outcomes in identifying wickets but are not able to identify actual examples altogether. The Others class has achieved the highest precision and recall values of 0.86 and 0.94, respectively. Macro and weighted average scores around 0.80 support that there is uniform performance on various classes, indicating that attention-based fusion was successful in enhancing multi-action recognition in various situations.

7.2.4 Model 4 : Late Fusion on C3D + CLAP

7.4 shows Confusion matrix of the late fusion model with C3D and CLAP: It is evident from the confusion matrix that the diagonal elements dominate, and most of the samples from

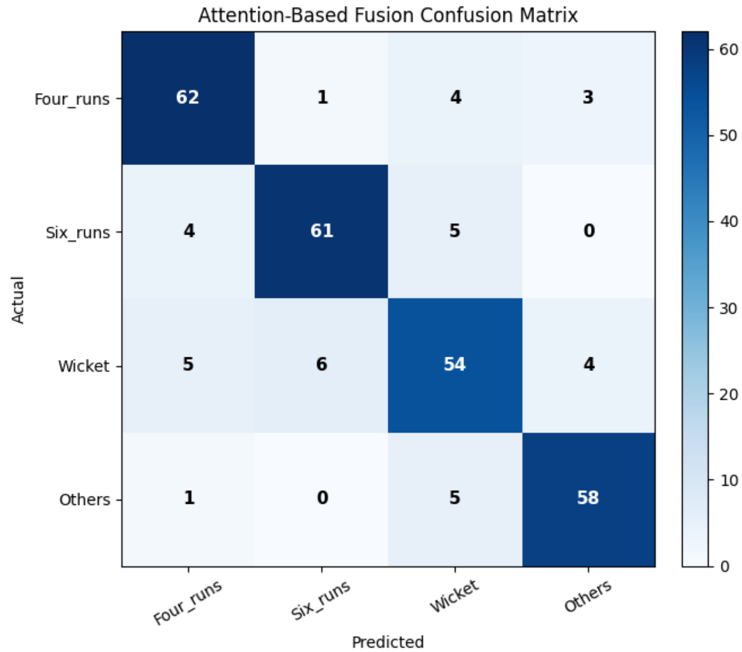

Figure 7.3: Confusion Matrix of Model 3 on Attention Fusion Mode

Table 7.5: Classification Report of Model 3 on Attention Fusion

Class	Precision	Recall	F1-score
Four_runs	0.68	0.91	0.78
Six_runs	0.81	0.81	0.81
Wicket	0.95	0.54	0.69
Others	0.86	0.94	0.90

all four action categories are classified correctly by the late fusion model. Six runs has the maximum correct classification with 32 out of 34 samples, and this is quite discriminative. Others is also quite reliable with 29 correct classifications, which indicates that there is negligible overlap with other classes. Confusion is mostly noticed for the Wicket class, where 6 samples are mistakenly classified under Six runs, implying that there is similarity in their motion dynamics and audio patterns. Similarly, only a few samples of Four runs are mistakenly classified under Others and Six runs, but this is still negligible.

Table 7.6 The classification report further gives a quantitative measure of performance by the model, with an overall accuracy of 87.31% and balanced macro and weighted F1-scores of 0.8734 and 0.8724, respectively. The Others class has the highest F1-score of 0.9206, influenced by strong precision of 0.9062 and recall of 0.9355. This indicates that this class is predicted reliably and consistently. For the six runs, a high recall of 0.9412 shows the ability of the model to identify this class with a small number of false negatives. Four runs

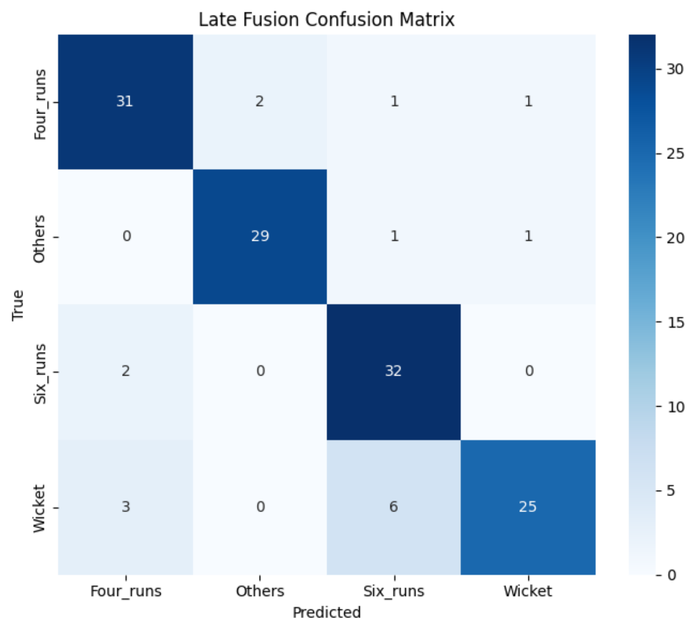

Figure 7.4: Confusion Matrix of Model 4 on Late Fusion Mode

has balanced precision and recall values, hence its F1-score is steady at 0.8696. Wicket has lower recall, 0.7647, hence some of its instances were missed despite the class having high precision of 0.9286. This proves that though the model predicts the Wicket class well, it would benefit from more training data or better feature presentation for this class.

Table 7.6: Classification Report of Model 4 on Late Fusion

Class	**Precision**	**Recall**	**F1-score**
Four_runs	0.85	0.85	0.86
Six_runs	0.93	0.93	0.92
Wicket	0.94	0.94	0.86
Others	0.92	0.76	0.83

7.2.5 Model 5 : Attention Fusion on C3D + MFCC

Fig. 7.5 From the confusion matrix, the efficiency of the attention-based fusion model is demonstrated with a large dominance on the diagonal, indicating the accurate predictions made by the system. All the wicket samples are predicted correctly without any confusion, further establishing the robustness of the system for that particular class. There is very little confusion in the Others class as well, with correct predictions for most of them. Where confusion is observed, it is between Four runs and Six runs with several samples for Four runs

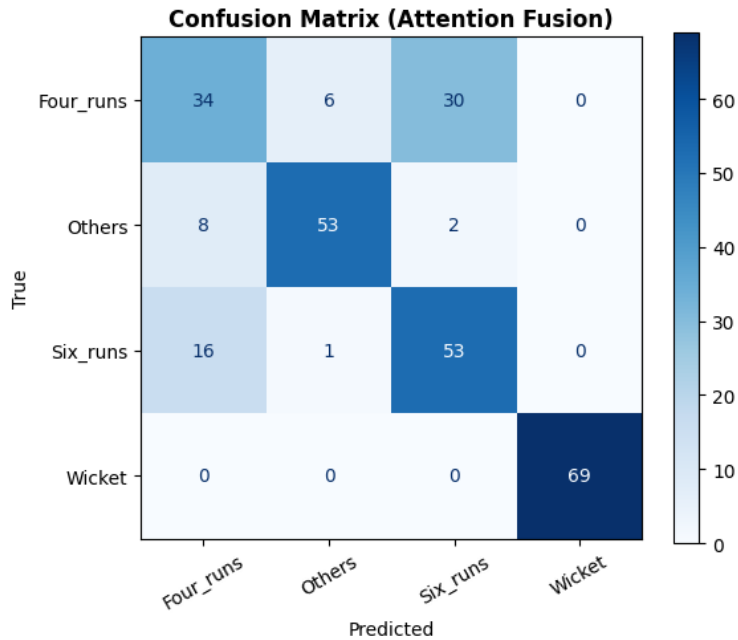

Figure 7.5: Confusion Matrix of Model 5 on Attention Fusion Mode

being predicted incorrectly as Six runs and vice versa. Although there is confusion between these two classes, most predictions are still correct, further establishing that mistakes lie between semantically similar instances rather than random events.

From Table 7.7 The classification report in Table 7.7 shows a total accuracy of 77%, along with a macro F1-score of 0.77 and a weighted F1-score of 0.77 for the attention-based fusion model. The Wicket class has a perfect classification with a precision of 100%, a recall of 100%, and a perfect F1-score of 1, which clearly verifies that the model performs well in identifying wicket events using their extremely distinctive audio-visual patterns. The Others classification had a great F1-score of 0.86, meaning that it effectively distinguishes non-highlight events as non-scoring actions. The Four runs and Six runs classification had a relatively low F1-score of 0.53 and 0.68 respectively, which occurred due to a large overlap between these two similar events in terms of their visual and audio-visual patterns.

Table 7.7: Classification Report of Model 5 on Attention Fusion

Class	**Precision**	**Recall**	**F1-score**
Four_runs	0.59	0.49	0.53
Six_runs	0.88	0.84	0.86
Wicket	0.62	0.76	0.68
Others	1	1	1

7.2.6 Bimodal Attention Fusion Results

Table 7.8 describes that attention fusion in Bimodal was also attempted on all possible model combinations to learn dynamically weighted audio and visual modalities based on classification. Although attention fusion showed competitive accuracy and enhanced inter-class discriminability for various models, it failed to outperform the best accuracy of late fusion for all models. Hence, although attention fusion offers important insights regarding multimodal interaction, late fusion with C3D and CLAP is still the most accurate method. In

Table 7.8: Bimodal Attention Fusion Results

Model	**Visual Model**	**Audio Model**	**Accuracy (%)**
1	ResNet18	YAMNet	0.82
2	R3D_18	MFCC	0.77
3	ResNet18	VGGish	0.75
4	C3D	CLAP	0.79
5	C3D	MFCC	0.78

all combinations of models considered for testing, ResNet18 + YAMNet with attention-based fusion performed best in terms of accuracy and class uniformity in addition to robustness in other attention-based bimodal fusion methods for both actions in the cricket game that look alike visually. This model makes optimal use of both audio and visual features. Although the highest accuracy of 87.3% was attained by the C3D+CLAP model with the fusion done at a later stage, the ResNet18+YAMNet model using attention-based fusion was found to perform better in terms of robustness for all categories of actions, with better performance for all classes. After careful consideration of accuracy, confusion matrices, and effectiveness of fusion, it was concluded that the audiovisual method using attention was the best feasible solution. Thus, this method was chosen to proceed with.

Chapter 8

CONCLUSIONS AND FUTURE SCOPE

8.1 Conclusion

This work successfully implements an accurate audio-visual action recognition system using ResNet18 for visual features, YAMNet for audio features, and a DNN-based attention fusion method. The attention fusion method is demonstrated to be more effective than early fusion and late fusion methods as it assigns dynamic weights to the audio and visual modalities, resulting in accurate predictions. Experimental analysis shows that the proposed model consistently generates accurate predictions at approximately 86%, which is more accurate than other tested combinations. Results obtained in the form of a confusion matrix and classification report show good distribution without a significant imbalance in action class predictions. This system successfully distinguishes actions involving comparable auditory stimuli like Four Runs and Six Runs using semantic embeddings. End-to-end testing provides validation for robustness from raw video to action prediction. The proposed method is scalable, feasible, and accurate for sports action recognition in a multimodal environment.

8.2 Future Scope

The current approach can be extended by incorporating modeling of time related to the long-term dynamics of motion over video frames. The approach can also be extended by training on the full UCF101 dataset or on other large sports datasets. Real-time implementation of the approach can also be explored by reducing feature extraction and inference time. The attention part of the approach can also be improved by learning attention coefficients using multi-head attention. The approach can also be extended by incorporating localization of events for action detection on individual frames instead of clips. The approach can also be extended by using domain adaptation methods for adapting different camera angles and handling noise in the background. The approach can also be applied for sports analysis.

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Appendix A

Plagiarism Report

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Figure A.1: Plagiarism Report